GeoPops: An open-source package for generating geographically realistic synthetic populations

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GeoPops

- **Open-source package for generating geographically and demographically realistic synthetic populations for any US Census location using publicly available data**
- Same methodology as GREASYPOP-CO (One Health Trust) but wrapped in convenient Python package that can be pip installed
- Automatic data downloading
- Compatibility with recent Census data (developing)
- Compatibility with ABM software Starsim (IDM)

github.com/ACCIDDA/GeoPops

```
pip install geopops
```

WSC Presentation

1. GeoPops Methodology
2. Make a population of Maryland, USA
3. Run SIR model of generic upper respiratory infection with Starsim
4. Track outcomes by age group and county
5. Repeat with Geo-Synthetic-Pop

github.com/ACCIDDA/GeoPops/tutorials/WSC

Larger Project

- SARS-CoV-2 model with SEIRD structure and age-dependent transmissibility and susceptibility
- First wave of the pandemic in Maryland
- Run on populations made with GeoPops and two other generators
- Compare agents and households to Census data using FT^2
- Compare network statistics of each population
- Compare simulated disease outcomes to observed data

GeoPops: Agents and networks

Agent attributes
- Age
- Gender
- Race/ethnicity
- Worker status
- Student status
- School grade
- Commuter status
- Income category
- Workplace category
- Home CBG
- School CBG if applicable
- Work CBG if applicable
- GQ CBG if applicable

School Network
- List of students for each
- Size of grade levels
- Number of teachers
- CBG locations

Workplace Network
- Number of workers
- CBG if inside population area

Household Network
- List of agents for each
- CBG locations

Group quarters (GQ) Network
- Number of residents
- Number of staff
- CBG locations

- Common network structure when modeling respiratory infections
- Few open-source methods for making this and fewer with spatial data

Step 1: Generating agents and households

Public Use Microdata Samples (PUMS)

Combinatorial optimization (CO)

American Community Survey (ACS)

- Subset of census survey responses
- Compositional details, e.g., the number of 5-member households in a CBG with school-aged children

- Demographic data at fine geographic scale
- Limited to marginal counts, e.g., the number of households in a CBG with 5 members and the number of households in a CBG with school-aged children

Alexander Y. Tulchinsky, Fardad Haghpanah, Alisa Hamilton, Nodar Kipshidze, and Eili Y. Klein. (2024). Generating geographically and economically realistic large-scale synthetic contact networks: A general method using publicly available data. https://arxiv.org/abs/2406.14698

Step 2: Assigning agents to schools and workplaces

	Data source	Data	Methods
Schools	National Center for Education Statistics (NCES)	Student and staff counts for each grade level in public schools	• Students assigned to closest school that offered required grade • Teachers assigned to school by selecting teachers from nearby CBGs
Workplaces	Longitudinal Employer-Household Dynamics (LEHD)	Origin-destination and workplace characteristics	• Iterative proportional fitting (IPF) to make origin-destination commute matrix • IPF to make industry by destination commute matrix for each origin CBG • Includes outside workers

Alexander Y. Tulchinsky, Fardad Haghpanah, Alisa Hamilton, Nodar Kipshidze, and Eili Y. Klein. (2024). Generating geographically and economically realistic large-scale synthetic contact networks: A general method using publicly available data. https://arxiv.org/abs/2406.14698

Step 3: Generating Contacts

Layer	Algorithm	Parameters
Household	All agents within a household are fully connected	• N/A
School	Stochastic Block Model within each school	• Assortativity: α=0.9 • Mean degree: K=12 • Blocks are grade levels (Pre-k – 12)
Workplace	Stochastic Block Model within each workplace	• Assortativity: α=0.9 • Mean degree: K=8 • Blocks are income levels (<40k or ≥ 40k)
Group quarters	Watts-Strogatz Model within each GQ facility	• Mean degree: K=12 • Rewiring probability: β=0.25

Alexander Y. Tulchinsky, Fardad Haghpanah, Alisa Hamilton, Nodar Kipshidze, and Eili Y. Klein. (2024). Generating geographically and economically realistic large-scale synthetic contact networks: A general method using publicly available data. https://arxiv.org/abs/2406.14698

Code example: MD, DC, northern VA

```
pars_geopops = {'path': "YOUR_OUTPUT_PATH",               # Set a folder where you want outut files to be stored
                'census_api_key': "YOUR_CENSUS_API_KEY", # Your Census API key
                'julia_env_path': "YOUR_JULIA_ENV_PATH", # Path to your Julia environment
                'main_year': 2019,                       # Year of data
                'geos': ["24",                           # State or county fips codes making up your population area
                         "11",
                         "51013","51059","51061", "51107","51153","51179","51510","51600","51610","51630","51683","51685"],
                'commute_states': ["24", "11", "51", "10", "42", "54"], # State fips of commute data to download
                'use_pums': ["24", "11", "51", "10", "42", "54", "25", "34", "37", "09"]} # State fips of PUMS data to download
geopops.WriteConfig(**pars_geopops) # Define parameters for pop generation in config.json
geopops.DownloadData()              # Download raw data files
geopops.ProcessData()               # Preprocess data
j = geopops.RunJulia()              # Run Julia portion
j.run_all()                         # Run all methods (CO(), SynthPop(), Export()) in  RunJulia() class
```

N ~9M. Several hours on my laptop, but you only have to run it once (a year)!

Accuracy

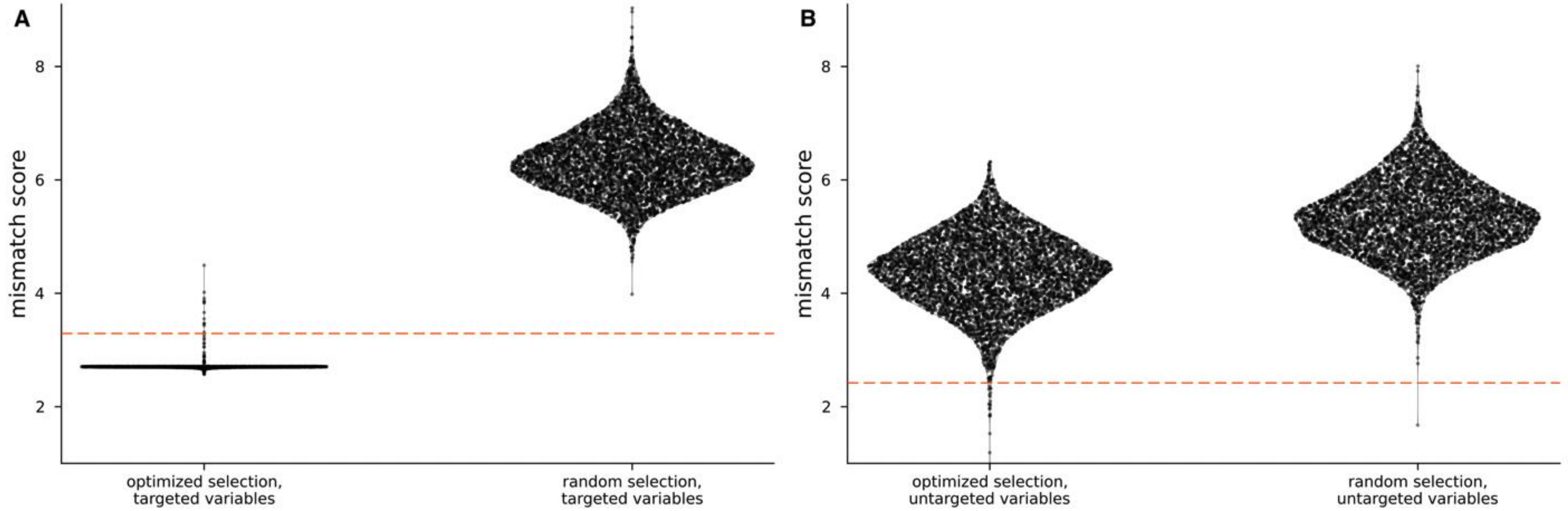
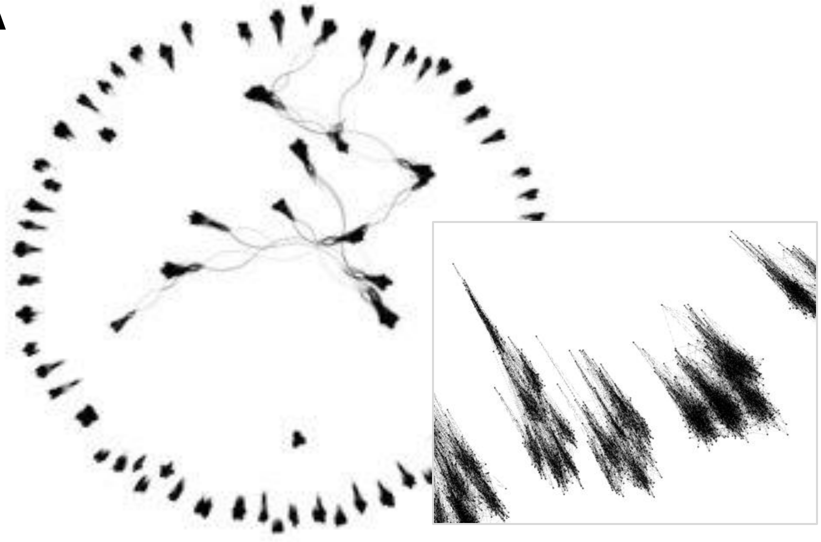


Fig 1. Fit between census data and households selected by combinatorial optimization for each census block group (CBG) in the Maryland-area synthetic population. (A) Comparison between an optimized selection of

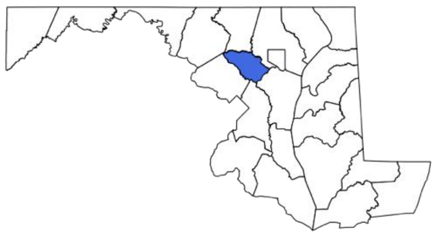
School Network Visualization (Howard County, MD)

A



B

p1	p2	beta
748	78	1
984	883	1
1702	115	1
1709	1521	1
1875	651	1



Visualization and table snippet of the school network. (A) Nodes are students and teachers clustered in grades within schools. (B) Same graph in Starsim tabular format where beta is the weight of the connection between person one and person two.

GeoPops with Starsim

Modules passed into Starsim to simulate disease transmission

GeoPops agents (ss.People)

SIR model (ss.Disease)
- Respiratory disease
- Infection duration ~7 days
- Transmission coefficient ~0.1
- 50-day simulation

GeoPops networks (ss.Network)

Subgroup Tracking (ss.Analyzer)
- Infections by age and CBG

Simple example with generic upper respiratory infection

Code Example

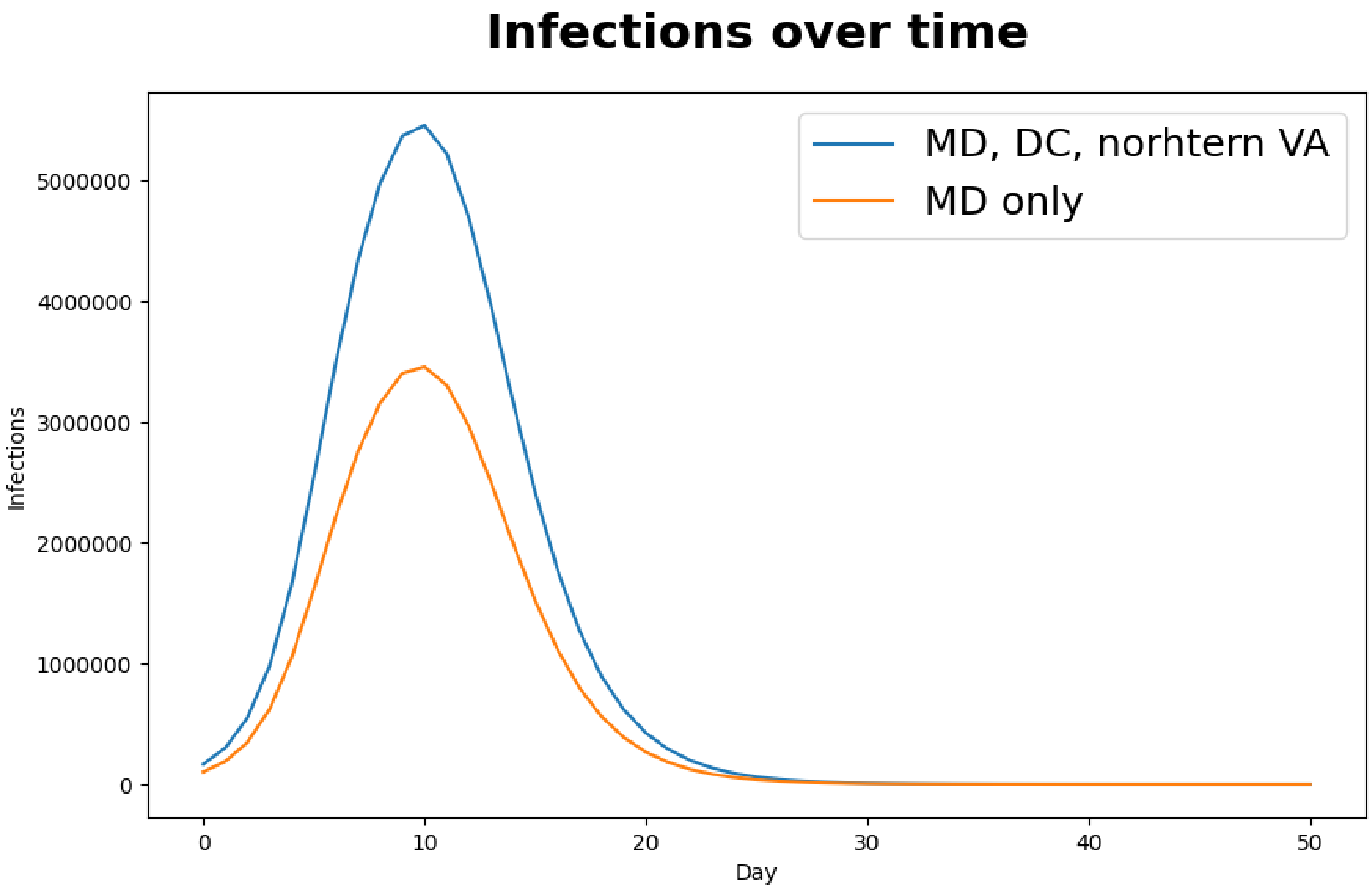
```
# Define epidemic parameters
pars_ss = sc.objdict(start = 0, stop=50, # Simulation duration
                     rand_seed = 123, # random seed
                     diseases = dict[str, str | float](type = 'sir', # type of disease
                                     init_prev = 0.01, # initial prevalence
                                     beta = 0.1,       # transmission rate
                                     dur_inf = 7,      # duration of infection
                                     ))
```

```
# Run SIR model with GeoPops people, networks, and subgroup tracking
sim_gp = ss.Sim(pars_ss, # Dictionary of SIR disease structure and parameters
                people=ppl_gp, # People object created with geopops.ForStarsim.People()
                networks=[h_gp,s_gp, w_gp, g_gp], # Created with geopops.ForStarsim.GPNetworks()
                analyzers=[age_tracking, county_tracking], # Created with geopops.ForStarsim.SubgroupTracking()
                ).run()
res_gp = sim_gp.results
```

✓ 12m 12.8s

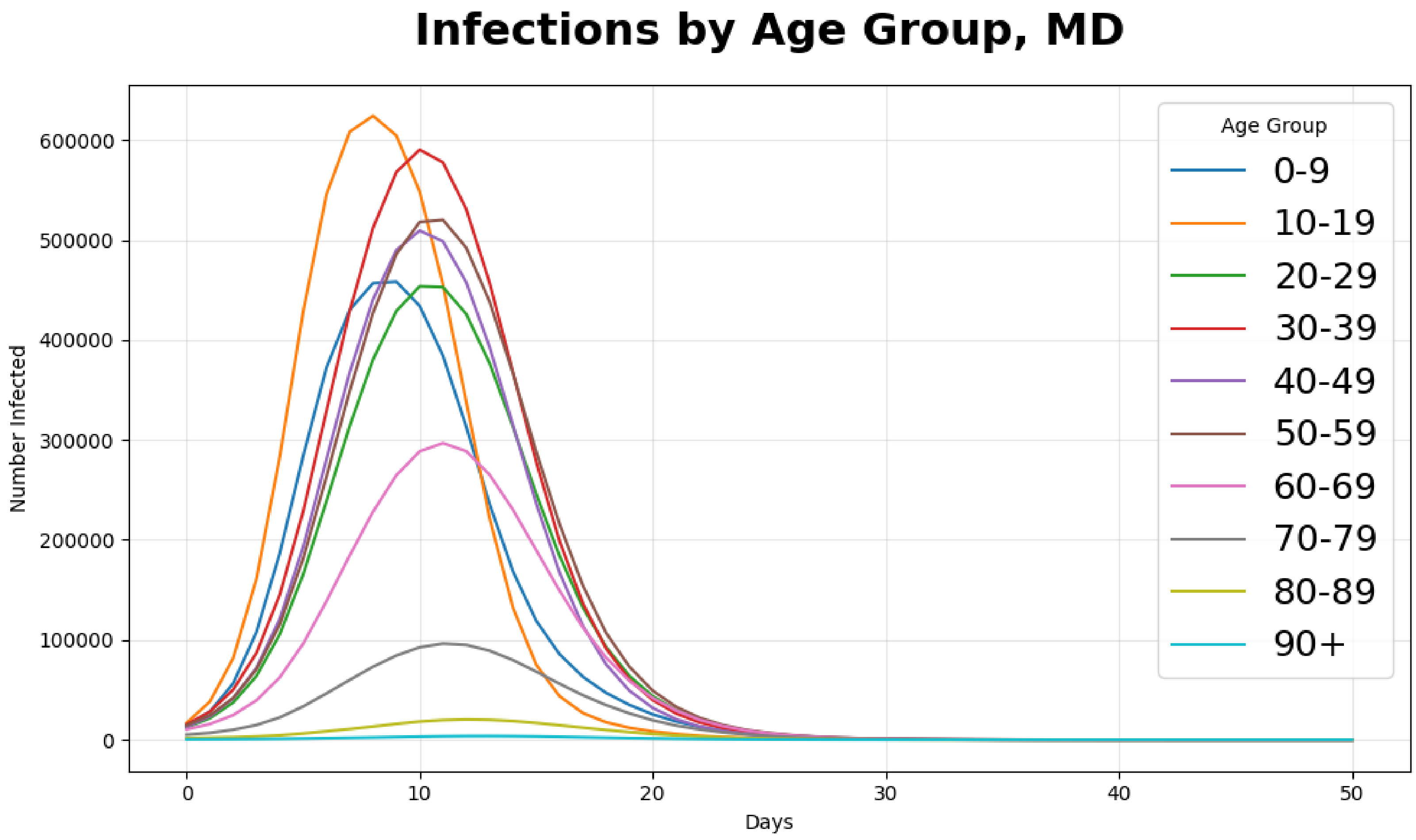
N ~9M

Infections

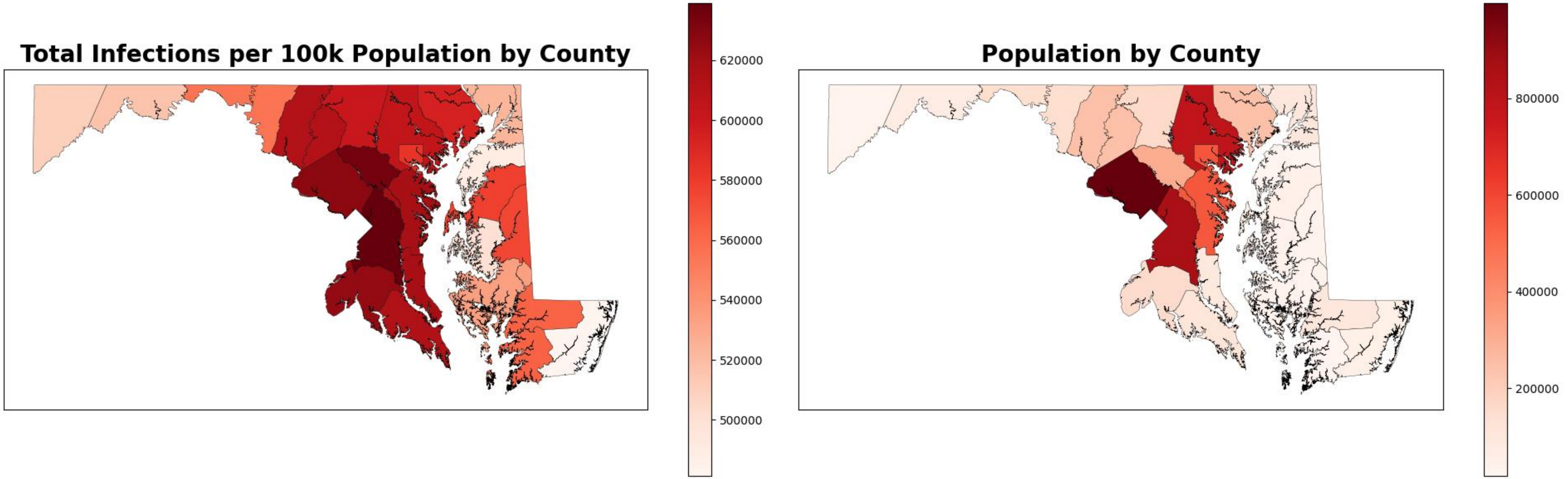


Infections over time
MD, DC, norhtern VA
MD only
5000000
4000000
3000000
2000000
1000000
0
Infections
0
10
20
30
40
50
Day

Outcome tracking by demographic subgroup



Outcome tracking by geographic subgroup



Why is this cool?!

Open-source, user-friendly framework for demographic and geographic subgroup tracking

- Don't have to build a model from scratch
- Can be used by state and county health departments for context-specific scenario modeling
- Seed infections to a specific geography or target interventions to specific demographic or geographic groups
- **Inform decisions about targeted interventions, which can lead to more efficient resource use and reductions in overall disease burden and disparities**
- There are several research groups already using Starsim to model respiratory diseases that can now use GeoPops as well

Actively maintained

Other Generators

Looking for…

- Open-source (downloadable datasets or reproducible code)
- Multiple demographic attributes
- CBG granularity (at least county)
- Home, school, workplace networks as graphs

Close, but limitations (for my purposes)…

- Geo-Synthetic-Pop (University at Buffalo)
- UrbanPop (Oak Ridge)
- SynthPops (Institute for Disease Modeling)
- FRED (RTI International)
- Epistorm (Indiana University)
- Others that aren't published yet

Geo-Synthetic-Pop

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nature > scientific data > data descriptors > article

Data Descriptor | Open access | Published: 07 November 2024

A Large-Scale Geographically Explicit Synthetic Population with Social Networks for the United States

Na Jiang, Fuzhen Yin, Boyu Wang & Andrew T. Crooks

https://doi.org/10.1038/s41597-024-03970-1

Pros

- Pre-built state populations
- All python
- Completely open-source

Cons

- Doesn’t work beyond 2019, Census boundaries changed
- No longer maintained (last activity over a year ago)
- State populations do not include out-of-state commuters in workplaces
- Fewer demographic attributes (only age, sex, and urbanicity)

UrbanPop

ELSEVIER

Applied Geography
Volume 151, February 2023, 102844

UrbanPop: A spatial microsimulation framework for exploring demographic influences on human dynamics

Joseph Tuccillo [a], Robert Stewart [a], Amy Rose [a], Nathan Trombley [b], Jessica Moehl [a], Nicholas Nagle [c], Budhendra Bhaduri [a]

https://doi.org/10.1016/j.apgeog.2022.102844

Pros

- All in python, actively maintained, responsive
- Really detailed re demographics
- Private and public schools with school names

Cons

- Need to be pretty good python programmer
- Code for assigning agents to schools and workplaces is not yet public
- Resulting dataset needs extra processing to be comparable to GeoPops
  - Agents within schools and workplaces are not connected with graph algorithms
  - Group quarters residents have household size of 1 and are not mapped to specific facilities
  - Workers are not mapped to specific workplaces within a CBG
  - No minimum constraints on school size

SynthPops

https://github.com/synthpops/synthpops

Pros

- Pre-built populations for all states
- All in python, open-source

Cons

- Doesn’t work with data beyond 2019
- No longer maintained
- Schools and workplaces do not have a geographical locations within the population area
- Fewer demographic attributes (only age and sex)
- Closed population (no out-of-state commuters)

FRED

Grefenstette *et al. BMC Public Health* 2013, **13**:940
http://www.biomedcentral.com/1471-2458/13/940

BMC Public Health

SOFTWARE Open Access

FRED (A Framework for Reconstructing Epidemic Dynamics): an open-source software system for modeling infectious diseases and control strategies using census-based populations

John J Grefenstette[1*], Shawn T Brown[2], Roni Rosenfeld[3], Jay DePasse[2], Nathan TB Stone[2], Phillip C Cooley[4], William D Wheaton[4], Alona Fyshe[3], David D Galloway[1], Anuroop Sriram[3], Hasan Guclu[1], Thomas Abraham[3] and Donald S Burke[1]

http://www.biomedcentral.com/1471-2458/13/940

Pros

- Pre-built populations

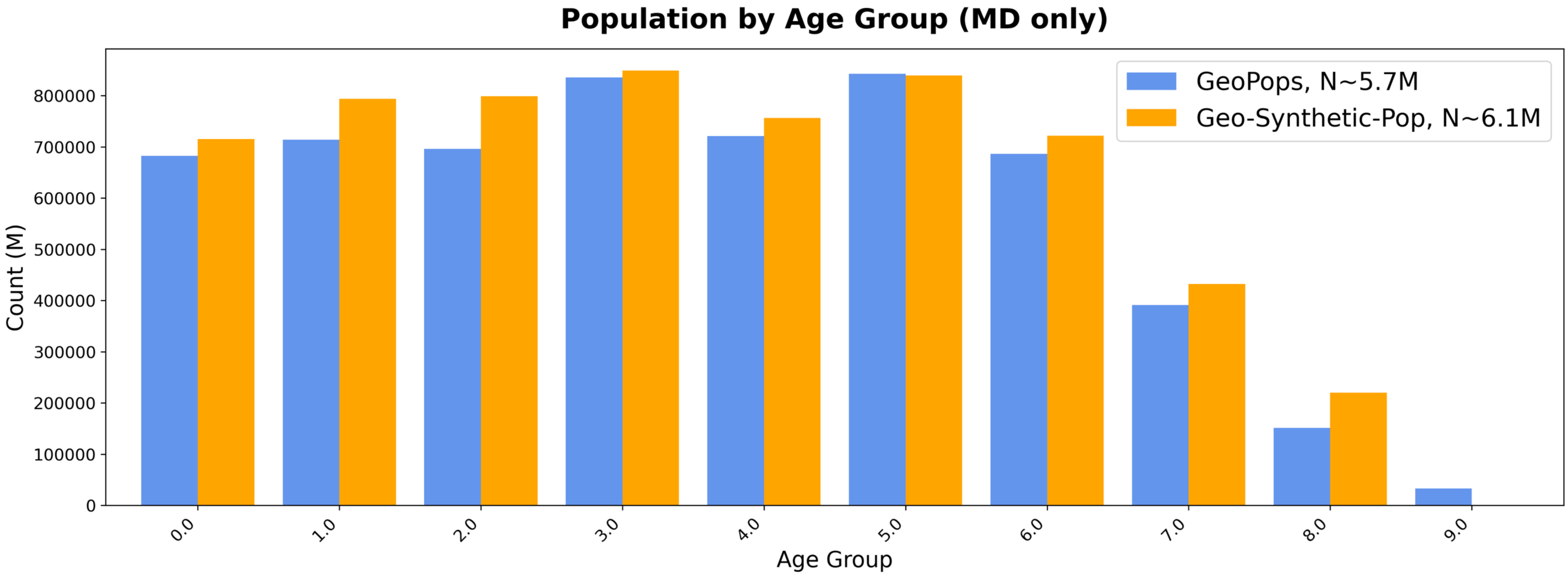
Cons

- C++
- Uses 2009 data
- No longer maintained

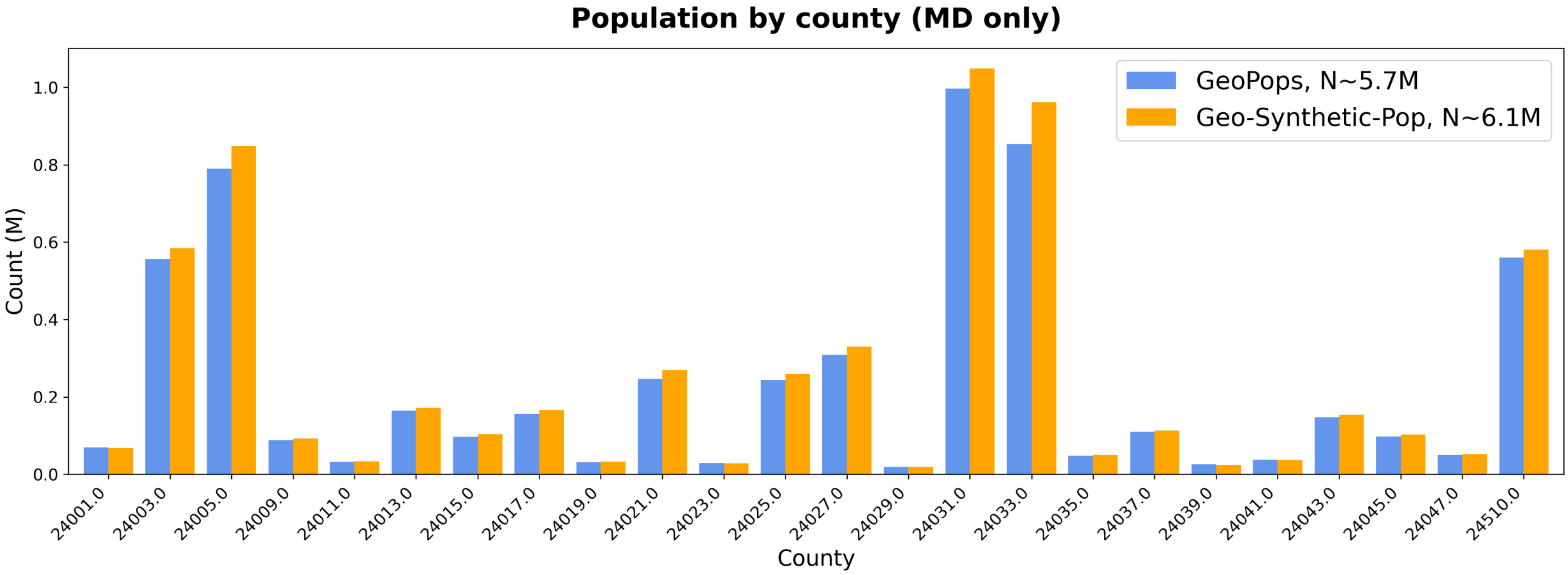
GeoPops vs Geo-Synthetic-Pop

	GeoPops	Geo-Synthetic-Pop
Agent and household generation	Combinatorial Optimization Age, sex, race/ethnicity, income, workplace category, CBG	Hierarchical Sampling Age, sex, urbanicity, Census Tract
School assignment	Assigned to closest school with capacity (NCES 2019)	Assigned to closest school with capacity (USEPA, Educational Institutions 2015)
Workplace assignment	LEHD origin-destination matrices and workplace characteristics (Iterative Proportional Fitting)	LEHD origin-destination matrices and workplace characteristics (Monte Carlo Sampling)

Population by Age Group (MD only)



Population by County (MD only)



GeoPops vs Geo-Synthetic-Pop
Network comparison

GeoPops = Blue (MD, DC, northern VA)

Geo-Synthetic-Pop = Orange (MD only)

Network	Algorithm	Mean degree (min, max)	No. Nodes	No. Edges
Home	Fully connected <5 Fully connected >5 Small-world	2.77 (1,19) 2.77 (1,11)	7,948,609 5,545,844	11,007,966 7,675,041
School	Stochastic Block Model <5 Fully connected >5 Small-world	12.01 (1,44) 5.20 (3,12)	1,762,561 1,014,435	10,583,502 2,636,664
Work	Stochastic Block Model <5 Fully connected >5 Small-world	7.55 (1,24) 3.53 (1,12)	4,557,404 1,814,870	17,193,880 3,201,197
Group Quarters	Small-world (Part of Home network)	12 (6,26)	200,873	1,205,292
Daycare	(Part of School network) <5 Fully connected >5 Small-world	5.20(1,13)	344,863	896,793

GeoPops vs Geo-Synthetic-Pop
Network comparison

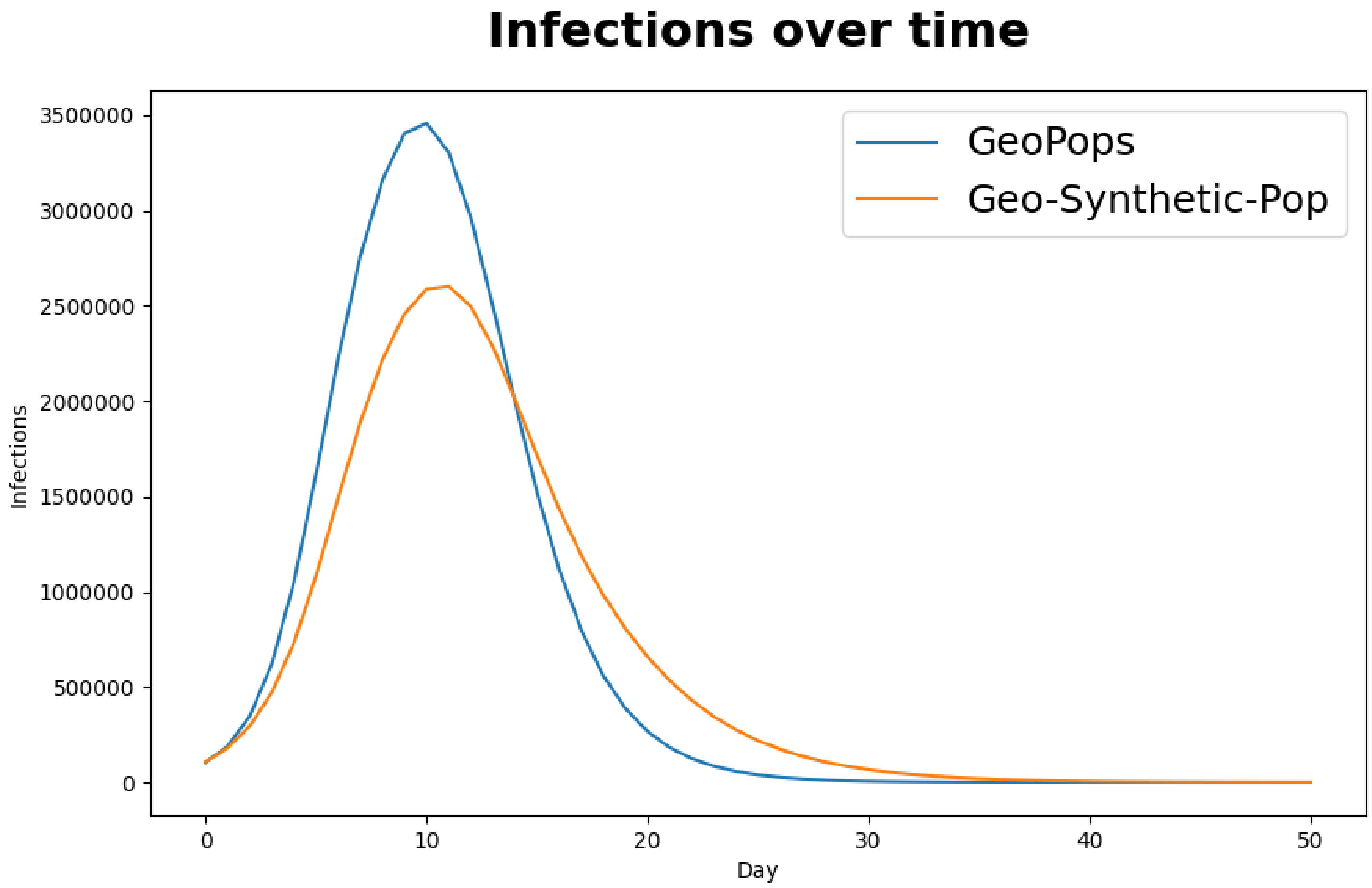
GeoPops

- Lets you specify commuter states and creates placeholder agents for out-of-state workers that are nodes with edges in the work network
- Group quarters residents are assigned to specific group quarters facilities and then connected within facilities using the Small-world algorithm
- Can be multiple group quarters facilities per CBG, each represented as a Small-world graph

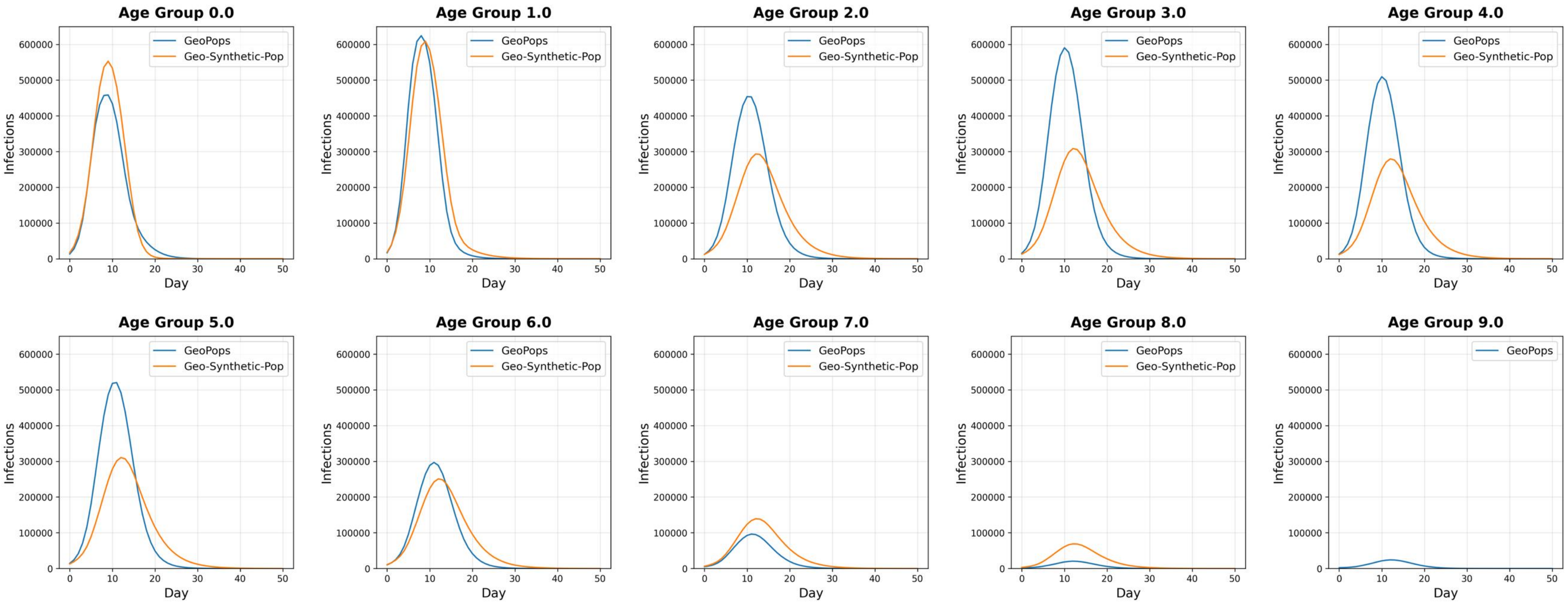
Geo-Synthetic-Pop

- Out-of-state commuters are not represented as nodes with edges in the work network
- Prisons, nursing homes, dorms, etc. are represented as one big “group-quarter household” per Tract, with the correct total population but no facility-level detail
- Only one Small-world graph per Tract

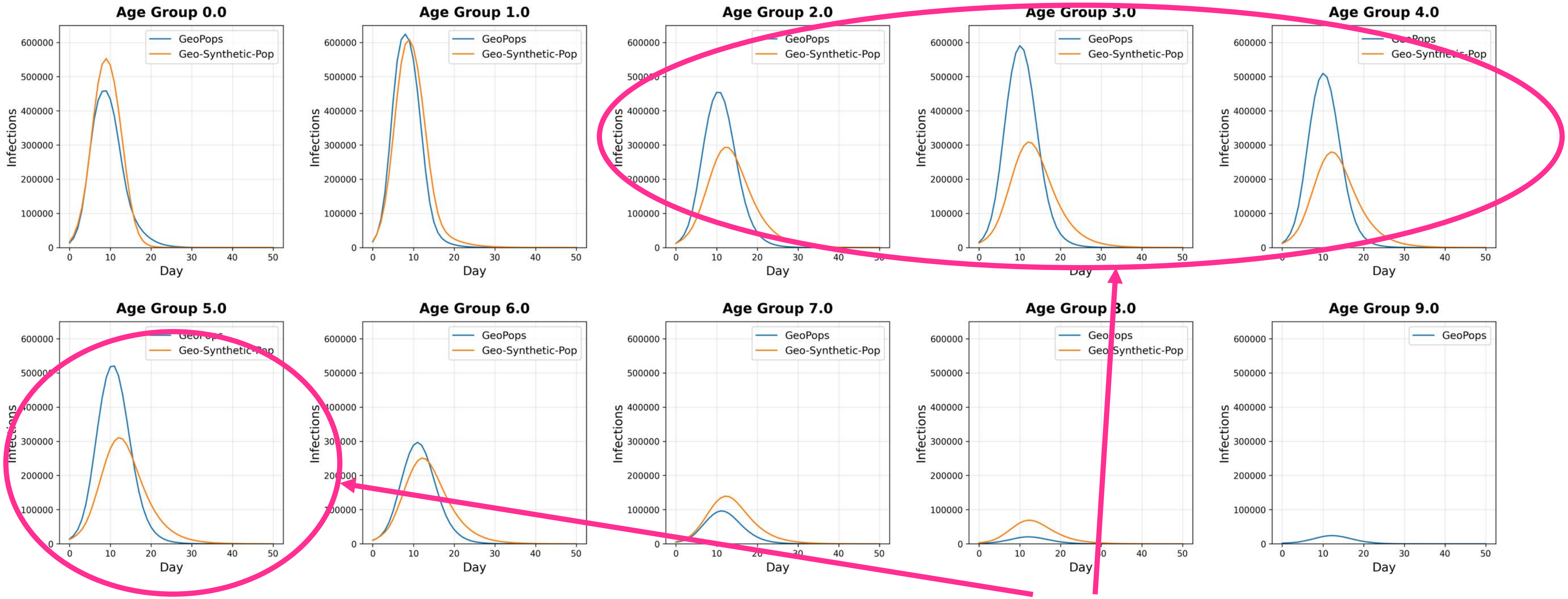
Comparison: Infections (MD only)



Comparison: Age Groups (MD only)

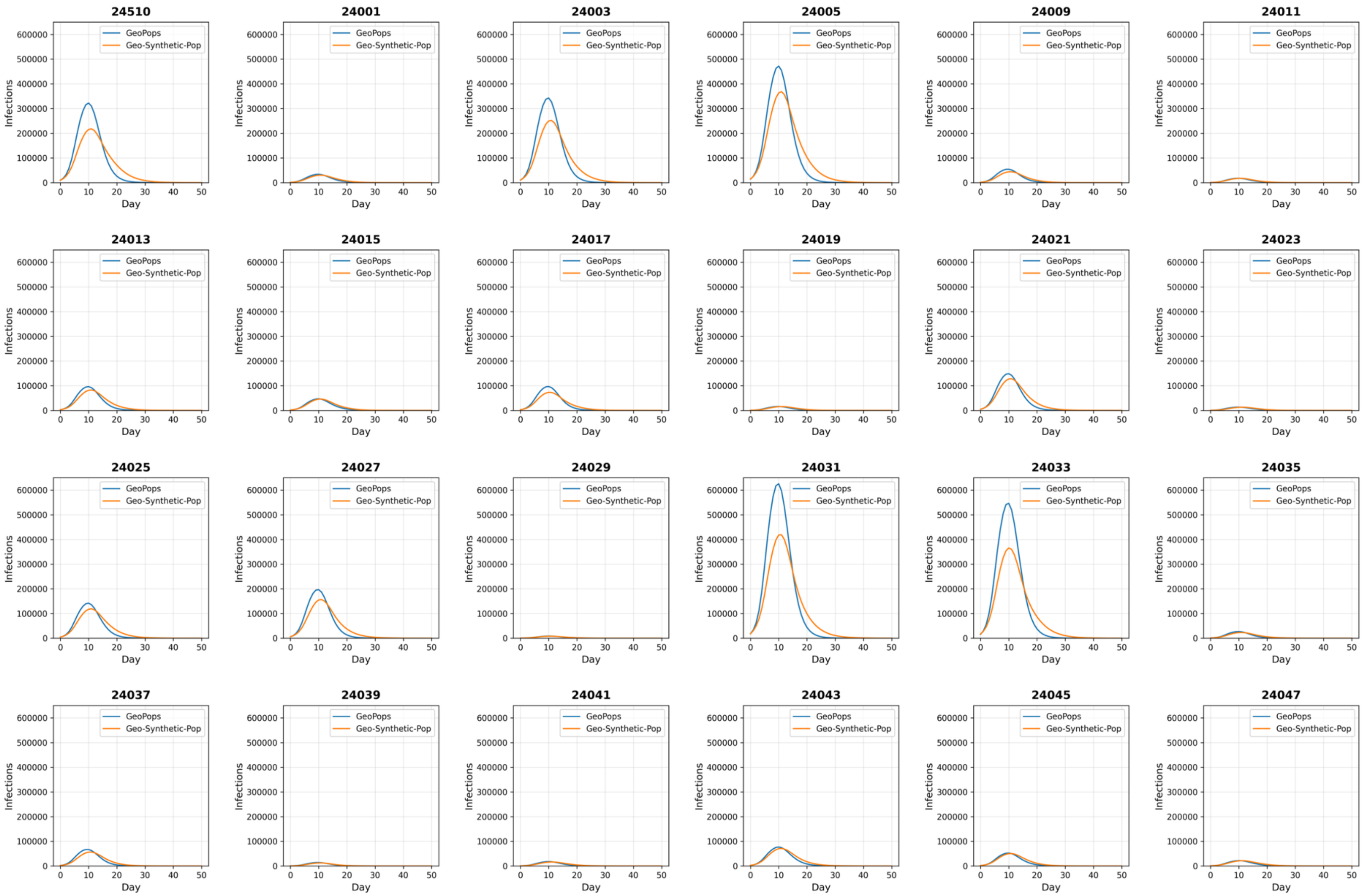


Comparison: Age Groups (MD only)



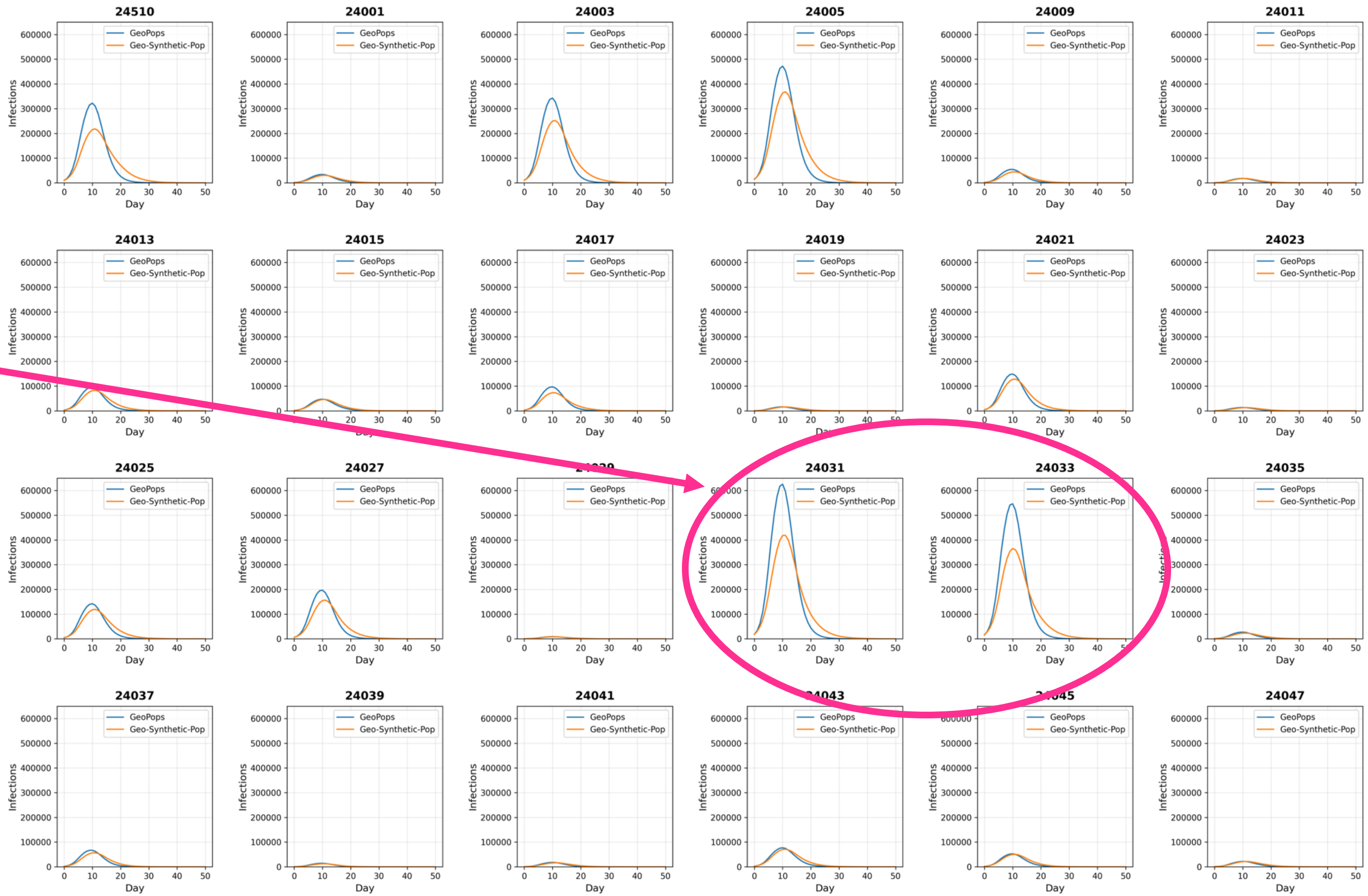
Working Age Groups

Comparison: Counties (MD only)



Comparison: Counties (MD only)

Montgomery and Prince George’s counties are right next to DC



To do…

- SARS-CoV-2 model with SEIRD structure and age-dependent transmissibility and susceptibility
- First wave of the pandemic in Maryland
- Run on populations made with GeoPops and two other generators
- Compare agents and households to Census data using FT^2
- Compare network statistics of each population
- Compare simulated disease outcomes to observed data

Questions

- Pip install geopops and try going through the tutorial
- Log issues on GitHub! Please be patient ☺
- Poster, slides, and Notebook tutorial on GitHub

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github.com/ACCIDDA/GeoPops/tutorials/MIDAS

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